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Digital Image Processing

Assignment - 1

Q.1 $\Rightarrow$ $m = 128$
 $n = 128$

No. of bands (channels) = 3

Bits per pixel $\Rightarrow 8 \text{ bit} = 1 \text{ byte}$

$\therefore$ Total memory required $\Rightarrow 128 \times 128 \times 3 \times 1 \text{ byte}$
 $\Rightarrow 48 \text{ KB}$

$\rightarrow$ In Grayscale image :-

$m = 128$

$n = 128$

No. of pixels (band) $\Rightarrow 1$

Bits per pixel $\Rightarrow 8 \text{ bits} = 1 \text{ byte}$

$\therefore$ Total memory required $\Rightarrow 128 \times 128 \times 1 \times 1 \Rightarrow 16 \text{ KB}$

$\therefore$ Memory reduction $\Rightarrow \frac{48 \text{ KB}}{16 \text{ KB}} \Rightarrow 3$

Q.3 $\Rightarrow$ Steps in Digital Image Processing :-

- (1) Image acquisition $\rightarrow$ capturing image using digital camera.
- (2) Preprocessing $\rightarrow$ removing noise etc.
- (3) Image enhancement $\Rightarrow$ Improving image quality for better interpretation.
- (4) Image restoration

- (5) Morphological processing
- (6) Segmentation
- (7) Object Recognition.
- (8) Representation and Description.
- (9) Image Compression
- (10) Color image processing.

Q.4 (a) Total size of packet $\Rightarrow 1+1+1 \Rightarrow 10$ bits

$\therefore$ Size of Image $\Rightarrow 1024 \times 1024 \times 10 \Rightarrow 10 \times 2^{20}$

$\therefore$ Size of baud Modem $\Rightarrow 750 \text{ K}$

$\therefore$ Time taken to pass over baud modem:-

$$\Rightarrow \frac{2^{20} \times 10}{750 \times 10^3} \Rightarrow \frac{1 \times 2^{20}}{75000} \text{ sec}$$

(b) Time taken to take over baud modem:-

$$\Rightarrow \frac{2^{20} \times (8+1+1)}{56 \times 1000} \Rightarrow \frac{10 \times 2^{20}}{56 \times 1000} \Rightarrow 3.12 \text{ minutes.}$$

Q.5

3	4	1	2	0	$\rightarrow Q$
0	1	0	4	2	$\uparrow$
2	2	3	1	4	
3	0	4	2	1	

$\downarrow p$

(a) Shortest 4-path:-

not - possible
Shortest 8-path:-

not - possible
Shortest m-path:-
not - possible.

(a)

$$V = \{0, 1, 2\}$$

$$(b) V = \{2, 3, 4\}$$

$\Rightarrow$ Shortest 4-path $\Rightarrow$ Not - possible

$\Rightarrow$ Shortest 8-path $\Rightarrow 4$

$$(3 \xrightarrow{p} 2 \rightarrow 3 \rightarrow 4 \rightarrow Q)$$

⇒ Shortest m-path ⇒ 7

(3 → 2 → 2 → 3 → 4 → 2 → 4 → 2)
P Q

Q.6 ⇒ Dimensions of Image ⇒ 100×100

each pixel can be either 0 or 1

∴ Total Distinct Image possible ⇒ $2^{100 \times 100}$

⇒ 2^{10000} images.

Q.7 ⇒ (a) Size of image in bits ⇒ $1024 \times 1024 \times 6$
⇒ 6×2^{20} bits

⇒ In bytes ⇒ $\frac{6 \times 2^{20}}{2^3} \Rightarrow 6 \times 2^{17}$ bytes.

⇒ In KiloBytes (KB) ⇒ $\frac{6 \times 2^{17}}{2^{10}} \Rightarrow 6 \times 2^7$ KB

∴ 1 GB ⇒ 2^{30} KB

∴ In GB ⇒ $\frac{6 \times 2^7}{2^{30}} \Rightarrow 6 \times 2^{-23}$ GB

(b) Storage ⇒ 6×2^{20} bits $\times 100$

Size of packet ⇒ 100 bits

Transfer rates ⇒ 128 K bits per second.

⇒ If image divided in packets, ~~then~~ ^{No. of} ~~size of~~
~~extra~~ packets ⇒ $\frac{6 \times 2^{20} \times 100}{100}$

extra bits per packet ⇒ $2^{20} \times 6 \times 28$

∴ Total bits ⇒ $\frac{600 \times 2^{20}}{100} + 6 \times 28 \times 2^{20} \Rightarrow 6 \times 2^{27}$
bits

$$\rightarrow \text{Time taken} \Rightarrow \frac{6 \times 2^{27}}{2^7 \times 1000} \Rightarrow 6 \times 2^{20} \times 10^{-3} \Rightarrow 6291.45 \text{ sec} \\ \Rightarrow 1.74 \text{ hrs.}$$

Q.8 $\Rightarrow$ Types of Image resolution :- To determine the quality and detail of an image :-

- (1) Spatial resolution :- No. of pixels per unit area in an image.
- (2) Gray-level resolution (Intensity resolution).
- (3) Temporal Resolution.
- (4) Spectral Resolution.

$\rightarrow$ Let phone A $\rightarrow$ higher spatial resolution.
 Let phone B $\rightarrow$ higher gray-level resolution.

$\rightarrow$ for sharp & detail images $\rightarrow$ phone A
 $\rightarrow$ for medical imaging, professional editing $\rightarrow$ phone B.

Q.10 $\Rightarrow$ (a) Euclidean distance $\Rightarrow \sqrt{(3-0)^2 + (3-0)^2} \Rightarrow \sqrt{18} \\ \Rightarrow 3\sqrt{2} \text{ Ans}$

(b) city-block distance $\Rightarrow |(3-0)| + |(3-0)| \Rightarrow 6 \text{ Ans}$

(c) Chess-board distance $\Rightarrow \max(|3-0|, |3-0|) \Rightarrow 3 \text{ Ans}$